

# Setting Up the Local Voice

How to install Kokoro, the free voice model ImageSkinForLLM uses to speak text, and check it works. It runs on the computer's processor (CPU): no graphics card, no account, no per-use cost. For Windows 11, macOS and Linux.

## What you will do

1. Check your Python version (This project with Kokoro needs Python 3.11 or 3.12).
2. Install the voice add-on into the project.
3. Run a test. The first run downloads the voice model once.

Time needed: about 10 to 15 minutes, mostly downloading. You need about 2 GB of free disk space and an internet connection for the first run.

**What this voice is.** Kokoro speaks with ready-made voices, not the voice of the person in the photo. Using the person's own voice is a later step, also free and on the CPU.

## Part 1. Check your Python version

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Open a terminal in the project folder (the folder with `README.md` ) and activate the virtual environment as described in the README.

### Windows 11, Command Prompt

```
.venv\Scripts\activate  
python --version
```

### Windows 11, PowerShell

```
.venv\Scripts\Activate.ps1  
python --version
```

### macOS / Linux

```
. .venv/bin/activate  
python --version
```

If it shows **3.11 or 3.12**, go to Part 2. If it shows 3.13 or later, make a new virtual environment with Python 3.12. On Windows, install Python 3.12 first if you don't have it: `winget install Python.Python.3.12` , then open a new terminal.

## Windows 11, Command Prompt

```
rmdir /s /q .venv
py -3.12 -m venv .venv
.venv\Scripts\activate
pip install -e ".[dev]"
```

## Windows 11, PowerShell

```
Remove-Item -Recurse -Force .venv
py -3.12 -m venv .venv
.venv\Scripts\Activate.ps1
pip install -e ".[dev]"
```

## macOS / Linux

```
rm -rf .venv
python3.12 -m venv .venv
. .venv/bin/activate
pip install -e ".[dev]"
```

# Part 2. Install the voice add-on

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With the virtual environment active, run this. It is the same on every system and installs Kokoro and PyTorch (about 1 GB):

```
pip install -e ".[voice]"
```

You do **not** need to install `espeak-ng` separately, even though Kokoro's own instructions mention it. Kokoro uses `espeak-ng` to pronounce words that aren't in its dictionary, and the command above installs a bundled copy for Windows, macOS and Linux (the `espeakng-loader` package).

**On a Linux server with no graphics card** (for example when the app is deployed in the cloud), first install the much smaller CPU-only PyTorch, then the line above:

```
pip install torch --index-url https://download.pytorch.org/whl/cpu
```

On Windows and macOS the normal install is already suitable for the CPU.

# Part 3. Test it

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Run this (the same on every system):

```
imageskin say -o hello.wav "Hello there, how are you today?"
```

The **first run** downloads the Kokoro model (about 330 MB) and a small English language file, so it can take a few minutes. Later runs reuse them and start in seconds.

It should print `Wrote <full path>\hello.wav (N.N seconds)` and word timings to `<full path>\hello.json`.

**Where the files go:** `hello.wav` and `hello.json` are created by this command in the folder you ran it from, which is the project folder if you followed Part 1. They are not part of the downloaded code. The printed message shows the full path. To save somewhere else, give a full path, for example `-o C:\Users\you\Desktop\hello.wav` (Windows) or `-o ~/Desktop/hello.wav` (macOS/Linux). To list them: `dir hello.*` (Command Prompt), `Get-ChildItem hello.*` (PowerShell) or `ls -l hello.*` (macOS/Linux).

Play the audio:

| System                     | Command                            |
|----------------------------|------------------------------------|
| Windows 11, Command Prompt | <code>start hello.wav</code>       |
| Windows 11, PowerShell     | <code>Invoke-Item hello.wav</code> |
| macOS                      | <code>afplay hello.wav</code>      |
| Linux                      | <code>aplay hello.wav</code>       |

`hello.json` lists each word with the time, in seconds, when it starts and ends. The app uses these times to highlight each word as it is spoken.

## How fast is it on your computer?

Look in the log lines the command prints for `"Spoke text"`. Its `real_time_factor` is how long speaking took divided by the length of the audio. **Below 1.0 means faster than real time**, which is what a live chat needs. The first run is slower because it loads the model.

## Part 4. Choose a voice

Add `--voice` and a voice name. The first letter is the accent (**a** = American, **b** = British); the second is **f** for female or **m** for male.

```
imageskin say --voice am_michael -o hello.wav "Hello there, how are you today?"
```

| American female                                                                                            | American male                                                              | British                                                                            |
|------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------|------------------------------------------------------------------------------------|
| <code>af_heart</code> (default), <code>af_bella</code> ,<br><code>af_nicole</code> , <code>af_sarah</code> | <code>am_michael</code> ,<br><code>am_fenrir</code> , <code>am_puck</code> | <code>bf_emma</code> , <code>bm_george</code> (spoken with American pronunciation) |

The full list is on the model's page: <https://huggingface.co/hexgrad/Kokoro-82M> (file VOICES.md). Each voice is downloaded the first time it is used.

## If something goes wrong

| The message mentions                                           | What to do                                                                                                                         |
|----------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------|
| Kokoro is not installed                                        | Run Part 2 with the virtual environment active.                                                                                    |
| Requires-Python or "no matching distribution" while installing | Your Python is too new. Follow Part 1 to use Python 3.12.                                                                          |
| could not load the Kokoro model                                | The first-run download failed. Check your internet connection, VPN or firewall (it downloads from huggingface.co), then try again. |
| Kokoro could not speak with voice                              | The voice name is misspelled or doesn't exist. Use one from Part 4.                                                                |
| A warning about <b>symlinks</b> on Windows                     | Harmless. The download still works; it just uses a little more disk space.                                                         |